

Coordinate–Face Reduction at the Coherent Möbius Gate:

Physical Zero–Face Absorption, Jacobi Near–Isometry, and Three–Modulus Energy Localization

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Abstract

The preceding product–slope analysis reduces an orbit–sliced prime–Möbius energy estimate to a weighted coherent multiplicative spectrum. Three possible obstructions remain entangled there: zero residue coordinates, nonconstant complementary weights, and the proper coordinate faces of a three–modulus completion. We separate them exactly.

At the endpoint

$$Q = X^{267/400+o(1)}, \quad J = X^{133/400+o(1)}, \quad T = X^{193/500+o(1)},$$

choose a prime $q \asymp J$ above the physical orbit support. The part of either orbit slice on which $q \mid m_\alpha j + h$ satisfies

$$\boxed{V_{L,q|N} + V_{R,q|N} \ll X^{o(1)}(1 + Q/q)Q^2 \ll X^{o(1)}Q^3/J.}$$

Thus every physical $s = 0$ or $u = 0$ residue face may be removed before finite–field completion at exactly the inherited gate scale. The same holds for any fixed number of orbit–scale primes.

On the regular part, the raw ultra residue vector G has $G(0) = 0$. Its punctured centering

$$\beta = \frac{1}{q-1} \sum_{U \neq 0} G(U), \quad g(0) = 0, \quad g(U) = G(U) - \beta \quad (U \neq 0)$$

is both additively centered and still vanishes at zero. If $f = F - q^{-1} \sum_D F(D)$, then for $a, s \neq 0$

$$\mathcal{C}_{a,s}(F, G) = \mathcal{C}_{a,s}(f, g) - \beta f(-h/a) + (q-1)\bar{F}\beta.$$

Hence the centered core has zero complete- s trace, while every centering correction is explicit and rank one in the complementary s -coordinate. For the actual dyadic ultra weight, the standard zero–free–region estimate gives

$$|\beta| \ll_A (U/q)(\log X)^{-A}.$$

Complete residue cycles of the complementary interval therefore cancel exactly in the core; only its incomplete boundary, the mean corrections, and alias weights remain.

We next diagonalize that nonconstant boundary. If $c = w *_\times g$, then the two–character transform is a shifted Mellin transform

$$C_\chi = \sum_{x \in \mathbb{F}_q^\times} c(x) \overline{\chi(x-h)}.$$

The nonprincipal Jacobi matrix $K_{\chi,\psi} = J(\psi, \bar{\chi})$ satisfies

$$\boxed{K^*K = KK^* = (q-1)^2I - q\mathbf{1}\mathbf{1}^*}.$$

Its singular values are $q-1$, with multiplicity $q-3$, and 1 , with multiplicity one. Equivalently,

$$\sum_{\chi \neq \mathbf{1}} |C_\chi|^2 = \sum_{\psi \neq \mathbf{1}} |\widehat{w}(\psi)\widehat{g}(\psi)|^2 - \frac{q}{(q-1)^2} \left| \sum_{\psi \neq \mathbf{1}} \widehat{w}(\psi)\widehat{g}(\psi)\psi(h) \right|^2.$$

Thus the Jacobi layer is near-isometric, not an automatic source of cancellation. Nevertheless, for an interval weight w , Pólya–Vinogradov saves the full uniform Cauchy factor in the unweighted character average. A sharp punctured character example shows that an arbitrary nonconstant coherent weight may still saturate that factor.

Finally, we give an exact calculus for all eight character faces of three pairwise coprime orbit–scale primes. The faces are orthogonal over one CRT period. Their squared masses, by codimension, are

$$J^{-3}, \quad J^{-2}, \quad J^{-1}, \quad 1.$$

The coherent union P of the seven proper faces obeys

$$\|P\|_\infty \ll J^{-1}, \quad \|P\|_2^2 \asymp J^{-1}, \quad \|P\|_1 \asymp 1.$$

Grouping a reciprocal dyadic block first by the integer determinant $F = su - N$ removes the apparent incomplete–box boundary: every face has total absolute mass at most $X^{o(1)}X/J^3$, while

$$\sum_{s,u} |P(su - N)|^2 \ll X^{o(1)}X/J^4 = o(1).$$

The full face, however, remains $1 - O(J^{-1})$ on every exact equality and retains the terminal Möbius layer. Closing the orbit gate still requires either absorption of the exposed punctured-centering mean corrections and principal outer boundary together with actual nonprincipal spectral overlap, or a target-coupled, coefficient-weighted shifted–fiber Bessel theorem for the unsplit raw physical output. We prove neither route, and claim no parity breach, Hardy–Littlewood asymptotic, or twin–prime result.

Keywords: Möbius correlation; multiplicative character; Jacobi sum; Chinese remainder theorem; orbit energy; parity boundary.

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1 Introduction

The preceding papers reduce one calibrated prime–Möbius residual to a specific orbit–sliced energy problem. TPC-25 shows, in an earlier one-sided shell, that physical coefficient provenance must be retained before taking absolute values [5]. TPC-35 identifies the complete product–slope spectrum and the critical three-modulus alias scale [6]. TPC-36 then proves that the actual static row mask has only $X^{o(1)}$ signed projective cost, and relocates the remaining difficulty to the weighted coherent functional

$$\frac{1}{q-1} \sum_{\chi} |\widehat{b}(\bar{\chi})|^2 \left| \sum_s w(s) \widehat{R}_s(\chi) \right|^2 \quad (1)$$

after incomplete ranges and integer aliases are restored [7]. The present paper separates three coordinate effects that are entangled in (1): physical zero residues, the nonconstant complementary-factor weight, and the proper faces of a three-modulus completion.

The inherited sufficient estimate is

$$\boxed{V_L + V_R \ll X^{o(1)} \frac{Q^3}{J}}. \quad (2)$$

The sign-blind exact-equality energy has the larger baseline

$$V_{\text{eq}}^{\text{abs}} \ll X^{o(1)} Q^3 J. \quad (3)$$

At the high- β endpoint,

$$Q = X^{267/400+o(1)}, \quad J = X^{133/400+o(1)}, \quad T = X^{193/500+o(1)}, \quad (4)$$

so the available ratio between (2) and (3) is exactly J^{-2} in energy. Equivalently, the allowance above the previously isolated Gram-copy diagonal $X^{o(1)} Q^2 J$ is

$$\frac{Q^3/J}{Q^2 J} = \frac{Q}{J^2} = X^{1/400+o(1)}. \quad (5)$$

There is no room for an untracked fixed power of X .

1.1 Zero coordinates are removed on the physical equality

Choose a prime $q \asymp J$ larger than the positive orbit support and between the physical d - and ℓ -scales. On the part of the actual packet satisfying

$$q \mid m_\alpha j + h, \quad (6)$$

a fixed orbit time selects only $O(1 + Q/q)$ row labels, whereas a fixed row label occurs at no more than one orbit time. This elementary two-sided incidence, combined with the inherited row ℓ^2 -mass, gives

$$\boxed{V_{L,\text{sing}(q)} + V_{R,\text{sing}(q)} \ll X^{o(1)} (1 + Q/q) Q^2 \ll X^{o(1)} \frac{Q^3}{J}}. \quad (7)$$

If the ultra divisor is opened as

$$su = m_\alpha j + h, \quad (8)$$

then, because q is prime, the event in (6) is precisely the union of the physical faces $s = 0 \pmod{q}$ and $u = 0 \pmod{q}$. Thus the zero coordinates are absorbed before replacing equality by congruence. The argument extends to any fixed number of orbit-scale primes.

This result does not discard the terminal factorization. On the regular support $q \nmid m_\alpha j + h$, the point $s = 1, u = m_\alpha j + h$ remains present and both coordinates are units.

1.2 Punctured centering isolates the complete trace

After physical zero-face excision, the raw ultra residue vector G satisfies $G(0) = 0$. Ordinary additive centering would generally create a nonzero value at the excluded residue. We instead define

$$\beta = \frac{1}{q-1} \sum_{U \neq 0} G(U), \quad g(0) = 0, \quad g(U) = G(U) - \beta \quad (U \neq 0). \quad (9)$$

If $f = F - q^{-1} \sum_D F(D)$, then for $a, s \neq 0$ the raw affine correlation satisfies the exact identity

$$\mathcal{C}_{a,s}(F, G) = \mathcal{C}_{a,s}(f, g) - \beta f(-h/a) + (q-1) \overline{F} \beta. \quad (10)$$

Every correction is explicit and rank one in the complementary s -coordinate. The punctured-centered core has the exact complete-cycle cancellation

$$\sum_{s \in \mathbb{F}_q^\times} R_s(a) = 0. \quad (11)$$

For the actual dyadic ultra coefficient, a standard one-Möbius estimate gives, for every fixed $A > 0$,

$$|\beta| \ll_A \frac{U}{q} (\log X)^{-A}. \quad (12)$$

Hence all complete residue cycles in the complementary interval vanish from the centered core; the incomplete boundary, the displayed mean corrections, and alias weights remain. The logarithmic saving in (12) is not promoted here to a fixed power saving.

1.3 The two-character layer is a Jacobi near-isometry

Let w be the nonconstant complementary-factor weight and let $c = w *_{\times} g$ on $\mathbb{F}_q^\times$. The coherent transform can be written as

$$C_\chi = \sum_{x \in \mathbb{F}_q^\times} c(x) \overline{\chi(x-h)}. \quad (13)$$

Taking a second Mellin transform produces the nonprincipal Jacobi matrix

$$K_{\chi, \psi} = J(\psi, \bar{\chi}), \quad \chi, \psi \neq \mathbf{1}. \quad (14)$$

We compute its Gram matrix exactly:

$$\boxed{K^* K = K K^* = (q-1)^2 I - q \mathbf{1} \mathbf{1}^*}. \quad (15)$$

Thus its singular values are $q-1$, with multiplicity $q-3$, and 1, with multiplicity one. The Jacobi layer is therefore a scaled near-isometry rather than a source of free square-root cancellation. The same calculation gives the exact energy identity

$$\begin{aligned} \sum_{\chi \neq \mathbf{1}} |C_\chi|^2 &= \sum_{\psi \neq \mathbf{1}} |\hat{w}(\psi) \hat{g}(\psi)|^2 \\ &\quad - \frac{q}{(q-1)^2} \left| \sum_{\psi \neq \mathbf{1}} \hat{w}(\psi) \hat{g}(\psi) \psi(h) \right|^2. \end{aligned} \quad (16)$$

When w is the residue-count weight of an interval, Pólya–Vinogradov bounds its nonprincipal Mellin coefficients and saves one full factor q relative to uniform Cauchy in the unweighted character average. That conclusion is deliberately not transferred to the outer physical coefficient direction: a punctured single-character example still attains the coherent Cauchy factor. For the nonprincipal outer spectrum, a sufficient hypothesis is a Mellin-flatness estimate for the physical convolution $b *_{\times} f$, or an equivalent bound for the resulting Möbius Hadamard energy. The principal outer boundary remains an explicit separate term. Neither that boundary estimate nor the nonprincipal hypothesis is proved.

1.4 Three-modulus proper faces localize, but the full face remains

For three pairwise coprime primes $q_i \asymp J$, we decompose the CRT delta into its eight principal/nonprincipal coordinate faces. They are orthogonal over one CRT period, with squared masses

$$J^{-3}, \quad J^{-2}, \quad J^{-1}, \quad 1 \quad (17)$$

according to the number of nonprincipal coordinates. More strongly, the coherent sum P of all seven proper faces satisfies

$$\|P\|_\infty \ll J^{-1}, \quad \|P\|_2^2 \asymp J^{-1}, \quad \|P\|_1 \asymp 1. \quad (18)$$

Grouping first by the integer determinant $F = su - N$ removes an apparent incomplete-box boundary and preserves these exact period norms up to the critical multiplicity

$$1 + \frac{X}{q_1 q_2 q_3} = X^{1/400+o(1)}. \quad (19)$$

This is a fiberwise statement. It does not commute automatically with the coherent row and divisor sums in the physical energy.

On the genuine equality $F = 0$, the proper union has amplitude $O(J^{-1})$. Therefore its contribution closes directly from the sign-blind baseline:

$$V_{\text{eq,proper}} \ll X^{o(1)} J^{-2} Q^3 J = X^{o(1)} \frac{Q^3}{J}. \quad (20)$$

The full three-coordinate face has amplitude $1 - O(J^{-1})$ on the same equality and retains the terminal Möbius layer. An artificial bounded coefficient can also align with the ℓ^1 , rather than the ℓ^2 , size of the proper face. Consequently the periodic quadratic calculation is not a coefficient-free proof of (2).

1.5 Claim boundary and organization

The unconditional advances are the physical absorption (7), the exact punctured-centering and Jacobi identities, the orthogonal CRT face calculus, determinant-first periodization, and proper-face closure on the exact equality. The remaining alternatives are either separate absorption of the exposed punctured-centering mean corrections and principal outer boundary together with an actual nonprincipal Mellin-overlap theorem, or a target-coupled, coefficient-weighted shifted-fiber Bessel estimate for the unsplit raw physical output. We establish neither route. In particular, this paper proves no estimate for the full regular coherent face, no affine Chowla theorem, no parity breach, no positivity or prime-pair main term, no Hardy–Littlewood asymptotic, and no twin-prime theorem.

Section 2 fixes the physical object and the normalizations. Section 3 removes the physical zero coordinates. Section 4 proves the punctured identities and complete-cycle cancellation. Section 5 gives the exact weighted Jacobi spectrum. Section 6 develops the three-modulus face calculus and determinant-first reassembly. The final sections state the remaining gate, certificate scope, and formal nonclaims.

2 The inherited physical and coherent interface

We restate only the inputs used below. They come from the actual source-compatible packet and the reductions recorded in TPC-25, TPC-35, and TPC-36 [5–7]. Finite-field identities proved later in this paper are unconditional algebraic statements, but their application to the prime–Möbius packet is always made through the interface in this section.

2.1 Rows, orbit, and ultra increment

Fix $h \in \mathbb{Z} \setminus \{0\}$, put $H = \text{rad } |h|$, and work on one dyadic packet with

$$Q = LD, \quad QJ \asymp X, \quad R \leq T < U_0 \asymp X. \quad (21)$$

A physical row is

$$\alpha = (\ell_\alpha, d_\alpha), \quad m_\alpha = \ell_\alpha d_\alpha \asymp Q, \quad (22)$$

where $\ell_\alpha \asymp L$ is prime, $d_\alpha \asymp D$ is squarefree, and $(d_\alpha, H) = 1$. The source separation makes the integer m_α determine its row uniquely. There are $O(Q)$ rows. For the two physical coefficient systems,

$$\sum_{\alpha} |\gamma_{\alpha}^{(i)}| \ll X^{o(1)} Q, \quad (23)$$

$$\sum_{\alpha} |\gamma_{\alpha}^{(i)}|^2 \ll X^{o(1)} Q, \quad i = 1, 2. \quad (24)$$

These estimates retain the Möbius/source provenance used in the preceding papers; they are not hypotheses on an arbitrary Q -entry vector.

The orbit variable j lies in a positive interval $\mathcal{J}$ containing $O(J)$ integers, and

$$N_{\alpha}(j) := m_{\alpha}j + h \asymp X. \quad (25)$$

The terminal range is chosen so that

$$T < \min_{\alpha, j} N_{\alpha}(j) \leq \max_{\alpha, j} N_{\alpha}(j) < U_0. \quad (26)$$

Thus the terminal divisor $u = N_{\alpha}(j)$ is present in the ultra increment

$$C_m(j) = \sum_{\substack{T < u \leq U_0 \\ u | m_j + h}} -\mu(u) \log u, \quad |C_m(j)| \ll X^{o(1)}. \quad (27)$$

On the physical support every divisor $u \mid N_{\alpha}(j)$ obeys

$$(u, m_{\alpha}jH) = 1. \quad (28)$$

The actual multiplier has the form

$$\mathfrak{A}_{\alpha, \gamma}^{\text{act}}(j) = \mathfrak{m}_{\text{act}}(\alpha, \gamma) \Xi_{\alpha, \gamma}(j) \mathcal{W}_{\alpha, \gamma}(j/J), \quad (29)$$

where all displayed periodic and smooth factors are bounded and

$$\mathfrak{m}_{\text{act}}(\alpha, \gamma) = \mathbf{1}_{\ell_{\alpha} \neq \ell_{\gamma}} \mathbf{1}_{|m_{\alpha} - m_{\gamma}| > \Delta} \mathbf{1}_{(d_{\alpha}, d_{\gamma}) \leq G} \quad (30)$$

for the inherited pruning scales $\Delta = QX^{-\kappa_0}$ and $G = X^{\kappa_0}$. TPC-36 proves that this static mask, together with the fixed orbit cells and smooth factors, has $X^{o(1)}$ signed product-ready reassembly cost. That theorem does not complete the integer ranges modulo q , nor does it estimate the resulting coherent divisor sum.

2.2 The two orbit energies and their gate

Define the left slice and its energy by

$$Y_{\gamma, j}^L := \sum_{\alpha} \gamma_{\alpha}^{(1)} \mathfrak{A}_{\alpha, \gamma}^{\text{act}}(j) C_{m_{\alpha}}(j), \quad (31)$$

$$V_L := \sum_{\gamma, j} |Y_{\gamma, j}^L|^2. \quad (32)$$

The right slice exchanges the two row systems:

$$Y_{\alpha, j}^R := \sum_{\gamma} \gamma_{\gamma}^{(2)} \mathfrak{A}_{\alpha, \gamma}^{\text{act}}(j) C_{m_{\gamma}}(j), \quad (33)$$

$$V_R := \sum_{\alpha,j} |Y_{\alpha,j}^R|^2. \quad (34)$$

The inherited scalar residual is controlled once

$$V_L + V_R \ll X^{o(1)} \frac{Q^3}{J} \quad (35)$$

is proved on every retained packet. This is a sufficient interface, not a conclusion imported into this paper. The complete input-copy diagonal has already been bounded by

$$V_{L,\Delta} + V_{R,\Delta} \ll X^{o(1)} Q^2 J. \quad (36)$$

Consequently the outstanding task is coefficient-specific and off-diagonal; replacing it by a maximum over all abstract row vectors is unnecessarily strong.

For later sign-blind comparisons, put

$$C_m^{\text{abs}}(j) := \sum_{\substack{T < u \leq U_0 \\ u | m j + h}} \log u. \quad (37)$$

The divisor bound gives $C_m^{\text{abs}}(j) \ll X^{o(1)}$. Define $Y^{L,\text{abs}}$ from (31) by replacing $\gamma_\alpha^{(1)}$, $\mathfrak{A}_{\alpha,\gamma}^{\text{act}}(j)$, and the ultra summands by their absolute values, and define $Y^{R,\text{abs}}$ symmetrically. Then the row ℓ^1 -bound, the $O(Q)$ opposite rows, and the $O(J)$ orbit times give

$$\boxed{V_{\text{eq}}^{\text{abs}} := \sum_{\gamma,j} |Y_{\gamma,j}^{L,\text{abs}}|^2 + \sum_{\alpha,j} |Y_{\alpha,j}^{R,\text{abs}}|^2 \ll X^{o(1)} Q^3 J.} \quad (38)$$

This estimate is deliberately crude: it records the exact-equality envelope after all Möbius signs have been erased. It is used only when an algebraic face has a uniform scalar amplitude on $su = N_\alpha(j)$. It is not evidence of cancellation and cannot by itself imply (35).

2.3 Reciprocal dyadic blocks and product slopes

Open one divisor in (27) and write

$$su = N_\alpha(j) = \ell_\alpha d_\alpha j + h. \quad (39)$$

Partition u into $O(\log X)$ reciprocal dyadic blocks $u \asymp U$, $U \geq T$, and attach the corresponding interval

$$s \asymp X/U. \quad (40)$$

On the full Cartesian support of each block, $su = O(X)$. After fixing the opposite row, one orbit residue cell, and one bounded projective layer from TPC-36, the integer core has the schematic product-ready form

$$\sum_{\ell,d,u,s} b(\ell) f(d) g(u) w(s) \mathbf{1}_{su=\ell jd+h}. \quad (41)$$

Here b is independent of j and s , and the physical ultra coefficient on a clean dyadic block is

$$g_{\text{phys}}(u) = -\mu(u) \log u \Phi_U(u) \quad (42)$$

with a controlled dyadic cutoff Φ_U . The weight w records the incomplete complementary interval and any alias multiplicity introduced later. Formula (41) is a topology and provenance statement; it is not a complete finite-field identity.

For a prime $q \nmid h$, the corresponding complete product-slope model is defined on $G_q = \mathbb{F}_q^\times$ by

$$R_s(a) := \sum_{D \in \mathbb{F}_q} f(D) g(s^{-1}(aD + h)), \quad (43)$$

$$(Tb)(j, s) := \sum_{\ell \in G_q} b(\ell) R_s(\ell j), \quad (44)$$

$$S_w(j) := \sum_{s \in G_q} w(s) (Tb)(j, s). \quad (45)$$

With the unnormalized multiplicative transforms

$$\widehat{b}(\chi) = \sum_{x \in G_q} b(x) \overline{\chi(x)}, \quad \widehat{R}_s(\chi) = \sum_{a \in G_q} R_s(a) \overline{\chi(a)}, \quad (46)$$

TPC-36 gives the exact coherent identity

$$\mathcal{G}_q(b, f, g; w) := \sum_{j \in G_q} |S_w(j)|^2 = \frac{1}{q-1} \sum_{\chi} |\widehat{b}(\bar{\chi})|^2 \left| \sum_{s \in G_q} w(s) \widehat{R}_s(\chi) \right|^2. \quad (47)$$

The right side is the actual finite-model coherent functional, not an upper bound. The square-function spectrum $\sum_{j,s} |(Tb)(j, s)|^2$ controls an average over independent s 's; it does not control (47) without a possible factor $q-1$. TPC-36 proves that this factor is sharp for arbitrary centered data.

2.4 Endpoint and alias ledger

At the endpoint used throughout this paper,

$$Q = X^{267/400+o(1)}, \quad J = X^{133/400+o(1)}, \quad T = X^{193/500+o(1)}, \quad (48)$$

and the source scales satisfy $D = o(J)$, $J = o(L)$, and $QJ = X^{1+o(1)}$. The identities relevant to the present ledger are

$$\frac{Q}{J^2} = X^{1/400+o(1)}, \quad \frac{X}{J^3} = X^{1/400+o(1)}, \quad (49)$$

$$\frac{T}{J} = X^{107/2000+o(1)}, \quad \frac{Q}{T} = X^{563/2000+o(1)}. \quad (50)$$

One modulus $q \asymp J$ leaves $O(1 + X/q) = O(Q)$ possible integer determinants in a block. Three pairwise coprime orbit-scale moduli leave

$$O\left(1 + \frac{X}{J^3}\right) = X^{1/400+o(1)} \quad (51)$$

joint aliases. This count matches the allowance above the diagonal, but it does not show that principal, proper-coordinate, or full-coordinate faces reassemble with the required sign.

The logical division used below is therefore strict:

- (i) the physical equality is used first to remove residue-zero coordinates;
- (ii) punctured centering and two-character analysis concern the regular complete finite model;
- (iii) the three-modulus face calculus controls exact periodic masses and determinant multiplicities; and
- (iv) any transfer of those quadratic masses through the coherent physical sums requires an additional coefficient-specific theorem.

None of the inherited inputs asserts that theorem. In particular, the setup supplies no affine Möbius cancellation uniform in a moving shift, no suppression of the full CRT face, and no parity-breaking estimate.

3 Physical excision of the zero coordinate faces

The complete product–slope spectrum is naturally formulated on $\mathbb{F}_q^\times$. Before completing any integer variable modulo q , we remove the physical row–time pairs on which a complementary factor can be zero modulo q . This order avoids confusing a coordinate face of a modular relaxation with cancellation in the Möbius coefficient.

3.1 The inherited orbit slice

We use only the following properties of the physical packet. The left and right row sets have $O(Q)$ elements. A left row α has a distinct integer label

$$m_\alpha = \ell_\alpha d_\alpha \asymp Q, \quad (52)$$

where the source label ℓ_α is prime, and a coefficient satisfying

$$\sum_\alpha |\gamma_\alpha^{(1)}|^2 \ll X^{o(1)} Q. \quad (53)$$

The symmetric estimate holds for the right coefficients. The orbit set $\mathcal{J}$ is contained in a positive interval of diameter $O(J)$, and every $j \in \mathcal{J}$ is $O(J)$. Put

$$N_\alpha(j) = m_\alpha j + h. \quad (54)$$

The actual multiplier $\mathfrak{A}_{\alpha,\gamma}^{\text{act}}(j)$ is bounded, and the ultra increment

$$C_m(j) = \sum_{\substack{T < u \leq U_0 \\ u | m_j + h}} -\mu(u) \log u \quad (55)$$

satisfies

$$|C_m(j)| \ll X^{o(1)} \quad (56)$$

uniformly on the packet. Thus the left slice and its energy are

$$Y_{\gamma,j}^L := \sum_\alpha \gamma_\alpha^{(1)} \mathfrak{A}_{\alpha,\gamma}^{\text{act}}(j) C_{m_\alpha}(j), \quad (57)$$

$$V_L := \sum_{\gamma,j} |Y_{\gamma,j}^L|^2. \quad (58)$$

The right objects are obtained by exchanging the two row systems.

At the high– β endpoint,

$$Q = X^{267/400+o(1)}, \quad J = X^{133/400+o(1)}, \quad T = X^{193/500+o(1)}, \quad (59)$$

and the inherited source scales give

$$D = o(J), \quad J = o(L). \quad (60)$$

Choose a fixed constant K larger than the constants implicit in the positive orbit support. Bertrand’s postulate supplies, for all sufficiently large X , a prime

$$KJ < q < 2KJ. \quad (61)$$

After enlarging K once, we may and do arrange that

$$\max \mathcal{J} < q, \quad \max_\alpha d_\alpha < q < \min_\alpha \ell_\alpha, \quad q \nmid h. \quad (62)$$

In particular, $q \nmid m_\alpha j h$ on the physical support.

3.2 One auxiliary prime

Split the ultra increment by the target residue:

$$C_m^{\text{sing}(q)}(j) := \mathbf{1}_{q|m_j+h} C_m(j), \quad (63)$$

$$C_m^{\text{reg}(q)}(j) := \mathbf{1}_{q \nmid m_j+h} C_m(j). \quad (64)$$

Let $Y^{L,\text{sing}(q)}$, $Y^{L,\text{reg}(q)}$, and their energies be defined from (57) by using the corresponding increment. Define the right versions symmetrically.

Theorem 3.1 (Physical absorption of one zero-coordinate face). *Under (52)–(62),*

$$\boxed{V_{L,\text{sing}(q)} + V_{R,\text{sing}(q)} \ll X^{o(1)} \left(1 + \frac{Q}{q}\right) Q^2.} \quad (65)$$

Consequently, at $q \asymp J$,

$$\boxed{V_{L,\text{sing}(q)} + V_{R,\text{sing}(q)} \ll X^{o(1)} \frac{Q^3}{J}.} \quad (66)$$

Proof. For fixed j , let

$$\mathcal{A}_q(j) = \{\alpha : q \mid m_\alpha j + h\}. \quad (67)$$

If $q \mid j$, this set is empty because $q \nmid h$. Otherwise m_α lies in one residue class modulo q . The labels are distinct and lie in an interval of length $O(Q)$, so

$$|\mathcal{A}_q(j)| \ll 1 + \frac{Q}{q}. \quad (68)$$

Conversely, for fixed α , the congruence $m_\alpha j \equiv -h \pmod{q}$ selects one class of $j \pmod{q}$. The orbit interval has diameter smaller than q , and hence

$$\#\{j \in \mathcal{J} : \alpha \in \mathcal{A}_q(j)\} \leq 1. \quad (69)$$

For one fixed opposite row γ , Cauchy–Schwarz, (56), and (68) give

$$\begin{aligned} \sum_j |Y_{\gamma,j}^{L,\text{sing}(q)}|^2 &\ll X^{o(1)} \left(1 + \frac{Q}{q}\right) \sum_j \sum_{\alpha \in \mathcal{A}_q(j)} |\gamma_\alpha^{(1)}|^2 \\ &\leq X^{o(1)} \left(1 + \frac{Q}{q}\right) \sum_\alpha |\gamma_\alpha^{(1)}|^2 \\ &\ll X^{o(1)} \left(1 + \frac{Q}{q}\right) Q. \end{aligned} \quad (70)$$

There are $O(Q)$ opposite rows. Summing (70) over γ proves the left part of (65); the right part is symmetric. Finally, $Q/q \rightarrow \infty$ and $q \asymp J$, proving (66). $\square$

The terminology in the theorem is literal. On opening an ultra divisor, write

$$su = N_\alpha(j). \quad (71)$$

For a prime q ,

$$q \mid N_\alpha(j) \iff q \mid s \text{ or } q \mid u. \quad (72)$$

Thus [theorem 3.1](#) absorbs simultaneously the $s = 0$ and $u = 0$ residue faces of the physical equality. It does not estimate these faces by first replacing the equality with a congruence.

3.3 A fixed finite family of moduli

The preceding argument is stable under the fixed number of primes used in a CRT realization. For every fixed r , the prime number theorem supplies r distinct primes in the same fixed dilate of J for all sufficiently large X ; the scale separation in (60) then gives (62) for each of them.

Corollary 3.2 (Fixed finite face excision). *Let $q_1, \dots, q_r \asymp J$ be distinct primes satisfying (62), where r is fixed independently of X , and put $P = \prod_{i=1}^r q_i$. Define*

$$C_m^{\text{sing}(P)}(j) := \mathbf{1}_{(mj+h,P) > 1} C_m(j). \quad (73)$$

Then

$$V_{L,\text{sing}(P)} + V_{R,\text{sing}(P)} \ll_r X^{o(1)} \frac{Q^3}{J}. \quad (74)$$

On the complementary row-time support, every factor s, u in (71) is a unit modulo every q_i .

Proof. Partition the event $(N_\alpha(j), P) > 1$ according to the first prime q_i dividing the target. Apply theorem 3.1 to each disjoint piece. The square of the sum of at most r pieces is at most r times the sum of their squares. Since r is fixed, this proves (74). The final assertion follows from $su = N_\alpha(j)$. $\square$

3.4 Reduction of the full energy

For one prime, the exact pointwise decomposition is

$$Y_{\gamma,j}^L = Y_{\gamma,j}^{L,\text{reg}(q)} + Y_{\gamma,j}^{L,\text{sing}(q)}. \quad (75)$$

Therefore

$$V_L + V_R \leq 2(V_{L,\text{reg}(q)} + V_{R,\text{reg}(q)}) + 2(V_{L,\text{sing}(q)} + V_{R,\text{sing}(q)}). \quad (76)$$

The same statement holds for the fixed finite family in corollary 3.2. Hence a bound of size $X^{o(1)}Q^3/J$ for the regular energy implies the same bound for the full energy. The theorem proved here supplies the singular term in this reduction; it does not supply the regular estimate.

At the endpoint,

$$\frac{Q}{q} = X^{67/200+o(1)}, \quad \frac{Q^3}{J} = X^{167/100+o(1)}. \quad (77)$$

In particular, the zero-coordinate faces fit the required ledger without using a sign estimate. The terminal factorization $s = 1, u = N_\alpha(j)$ remains in the regular part whenever $q \nmid N_\alpha(j)$; no terminal Möbius cancellation is asserted.

4 Punctured centering of the regular ultra coefficient

After section 3, the physical equality is restricted to $q \nmid N_\alpha(j)$. Hence both factors in $su = N_\alpha(j)$ are nonzero modulo q . We now center the completed ultra coefficient without reintroducing an artificial value at the zero residue.

4.1 An exact asymmetric centering identity

Let $q > 2$ be prime, put $G_q = \mathbb{F}_q^\times$, and fix $h \in G_q$. Let $F, G : \mathbb{F}_q \rightarrow \mathbb{C}$, where

$$G(0) = 0. \quad (78)$$

The function F is the raw residue aggregation of a d -weight, and G is the raw aggregation of the regular ultra weight. Define the ordinary additive centering of F by

$$\bar{F} = \frac{1}{q} \sum_{D \in \mathbb{F}_q} F(D), \quad f(D) = F(D) - \bar{F}. \quad (79)$$

For G , instead put

$$\beta = \frac{1}{q-1} \sum_{U \in G_q} G(U), \quad (80)$$

and define its punctured centering by

$$g(0) = 0, \quad g(U) = G(U) - \beta \quad (U \in G_q). \quad (81)$$

Then

$$\sum_D f(D) = \sum_U g(U) = 0, \quad g(0) = 0. \quad (82)$$

For $a, s \in G_q$, write

$$\mathcal{C}_{a,s}(A, B) := \sum_{D \in \mathbb{F}_q} A(D) B(s^{-1}(aD + h)). \quad (83)$$

Proposition 4.1 (Punctured-centering identity). *For every $a, s \in G_q$,*

$$\boxed{\mathcal{C}_{a,s}(F, G) = \mathcal{C}_{a,s}(f, g) - \beta f(-h/a) + (q-1)\bar{F}\beta.} \quad (84)$$

Proof. On $\mathbb{F}_q$,

$$F = f + \bar{F}, \quad G = g + \beta \mathbf{1}_{G_q}. \quad (85)$$

The affine map $D \mapsto s^{-1}(aD + h)$ is a permutation, so the constant part of F contributes

$$\bar{F} \sum_U G(U) = (q-1)\bar{F}\beta. \quad (86)$$

For the other correction, the indicator $\mathbf{1}_{G_q}(s^{-1}(aD + h))$ deletes exactly $D_0 = -h/a$. Since f is centered,

$$\sum_D f(D) \mathbf{1}_{aD+h \neq 0} = -f(D_0). \quad (87)$$

Combining the two identities proves (84). $\square$

The identity extends directly to the product-slope topology. Define

$$R_s^{\text{raw}}(a) := \mathcal{C}_{a,s}(F, G), \quad R_s(a) := \mathcal{C}_{a,s}(f, g), \quad (88)$$

$$(T^{\text{raw}}b)(j, s) := \sum_{\ell \in G_q} b(\ell) R_s^{\text{raw}}(\ell j), \quad (Tb)(j, s) := \sum_{\ell \in G_q} b(\ell) R_s(\ell j). \quad (89)$$

Put

$$B_0 = \sum_{\ell \in G_q} b(\ell), \quad H_f(j) = \sum_{\ell \in G_q} b(\ell) f\left(-\frac{h}{\ell j}\right). \quad (90)$$

Then [proposition 4.1](#) gives the pointwise formula

$$\boxed{(T^{\text{raw}}b)(j, s) = (Tb)(j, s) - \beta H_f(j) + (q-1)\bar{F}\beta B_0.} \quad (91)$$

For an arbitrary weight $w : G_q \rightarrow \mathbb{C}$, with $W_0 = \sum_s w(s)$,

$$\begin{aligned} \sum_s w(s) (T^{\text{raw}}b)(j, s) &= \sum_s w(s) (Tb)(j, s) \\ &\quad + W_0 \{-\beta H_f(j) + (q-1)\bar{F}\beta B_0\}. \end{aligned} \quad (92)$$

Thus every centering correction is explicit and contains the single scalar β . No smallness assumption on $\bar{F}$ is used in this algebraic statement.

4.2 The actual Möbius mean

On a reciprocal dyadic block, choose the canonical allocation in which the ultra factor remains

$$a_U(u) = -\mu(u) \log u \Phi_U(u), \quad (93)$$

where Φ_U is the indicator of $\{u : U < u \leq 2U, T < u \leq U_0\}$, or a bounded smooth/BV cutoff with controlled seminorms on that interval. On the regular physical support one may insert $\mathbf{1}_{q \nmid u}$ exactly. The raw residue aggregation is therefore

$$G_U(r) = \sum_{\substack{u \equiv r \\ q \nmid u}} \frac{(\text{mod } q)}{q} -\mu(u) \log u \Phi_U(u), \quad r \in \mathbb{F}_q, \quad (94)$$

and indeed $G_U(0) = 0$. Its punctured mean is

$$\beta_U = -\frac{1}{q-1} \sum_{\substack{u \geq 1 \\ q \nmid u}} \mu(u) \log u \Phi_U(u). \quad (95)$$

Proposition 4.2 (Uniform one-Möbius bound for the punctured mean). *Assume $q \asymp J$, $U \geq T$, and (59), and assume that the cutoff family in (93) has uniformly bounded smooth or BV seminorms. For every fixed $A > 0$,*

$$\boxed{|\beta_U| \ll_{A,\Phi} \frac{U}{q} (\log X)^{-A}.} \quad (96)$$

The estimate is uniform over the dyadic ultra blocks.

Proof. Let

$$M(x) = \sum_{n \leq x} \mu(n), \quad M_q(x) = \sum_{\substack{n \leq x \\ (n,q)=1}} \mu(n). \quad (97)$$

Separating the multiples of the prime q gives the exact recurrence

$$M_q(x) = \sum_{r \geq 0} M(x/q^r), \quad (98)$$

where the sum terminates. The standard zero-free-region form of the prime number theorem [3, Chapters 5 and 18] gives, for every fixed $B > 0$,

$$M(y) \ll_B y (\log y)^{-B} \quad (99)$$

when y is large. In (98), the terms after the first have total trivial size $O(x/q)$, which is smaller than $x(\log X)^{-B}$ because $q = X^{133/400+o(1)}$. Hence, uniformly for $x \asymp U$,

$$M_q(x) \ll_B U (\log X)^{-B}. \quad (100)$$

Partial summation, with B chosen beyond A and the fixed seminorm loss of Φ_U , yields

$$\sum_{q \nmid u} \mu(u) \log u \Phi_U(u) \ll_{A,\Phi} U (\log X)^{-A}. \quad (101)$$

Division by $q-1 \asymp q$ proves (96). □

The endpoint separation

$$\frac{T}{q} = X^{107/2000+o(1)} \quad (102)$$

also permits the same argument after writing a zero-residue sum as $u = qv$. More importantly, if $|\mathcal{S}_U| \ll X/U$, then

$$|\beta_U| |\mathcal{S}_U| \ll_A \frac{X}{q} (\log X)^{-A} \asymp Q (\log X)^{-A}. \quad (103)$$

This is an arbitrary logarithmic saving, not a fixed power of X . Even a Vinogradov–Korobov error has size $X^{-o(1)}$, so [proposition 4.2](#) alone does not pay the $X^{1/400}$ endpoint margin. Nor may its logarithmic saving be placed after an uncontrolled worst-case $X^{o(1)}$ absolute reassembly. The mean must be extracted while the ultra weight still has the clean form [\(93\)](#).

4.3 Exact cancellation of complete residue cycles

The punctured-centered core has no complete complementary-factor trace.

Proposition 4.3 (Complete-cycle annihilation). *For every $a, j \in G_q$,*

$$\sum_{s \in G_q} R_s(a) = 0, \quad \sum_{s \in G_q} (Tb)(j, s) = 0. \quad (104)$$

Proof. Fix a and put $D_0 = -h/a$. The term $D = D_0$ vanishes because $g(0) = 0$. For $D \neq D_0$, as s runs over G_q , the value $s^{-1}(aD + h)$ runs once over G_q . Its total contribution is zero because $\sum_{U \in G_q} g(U) = 0$. This proves the first identity; the second follows by summing in ℓ . $\square$

For an integer interval $\mathcal{S}$, define its nonzero residue lift count by

$$W_{\mathcal{S}}(r) := \#\{s \in \mathcal{S} : s \equiv r \pmod{q}\}, \quad r \in G_q. \quad (105)$$

Writing $|\mathcal{S}| = kq + r_0$, one has

$$W_{\mathcal{S}}(r) = k + \rho_{\mathcal{S}}(r), \quad \rho_{\mathcal{S}}(r) \in \{0, 1\}, \quad |\text{supp } \rho_{\mathcal{S}}| < q. \quad (106)$$

Consequently,

$$\sum_{r \in G_q} W_{\mathcal{S}}(r) (Tb)(j, r) = \sum_{r \in G_q} \rho_{\mathcal{S}}(r) (Tb)(j, r). \quad (107)$$

For the complementary interval $|\mathcal{S}_U| \asymp X/U$, the number of removed complete cycles satisfies

$$k_U \ll 1 + \frac{X}{Uq} \asymp 1 + \frac{Q}{U}. \quad (108)$$

It is $\asymp Q/U$ whenever the complementary interval contains at least a fixed positive number of complete residue cycles. In particular,

$$k_T = X^{563/2000+o(1)}, \quad (109)$$

whereas $k_U = 0$ once U exceeds a sufficiently large fixed multiple of Q .

Equation [\(107\)](#) concerns the core residue lift count. An alias selector or another nonconstant reassembly weight belongs to ρ -type weighted data and is not annihilated by this proposition.

4.4 Comparison with ordinary additive centering

If one instead uses

$$\overline{G} = \frac{1}{q} \sum_U G(U) = \frac{q-1}{q} \beta, \quad g^{\text{add}} = G - \overline{G}, \quad (110)$$

then

$$g^{\text{add}}(0) = -\frac{q-1}{q} \beta \quad (111)$$

despite the raw identity $G(0) = 0$. In fact,

$$g^{\text{add}} = g + \frac{\beta}{q} - \beta \mathbf{1}_{\{0\}}. \quad (112)$$

The complete-trace formula for ordinary centering becomes

$$\sum_{s \in G_q} \mathcal{C}_{a,s}(f, g^{\text{add}}) = -(q-1)\beta f(-h/a). \quad (113)$$

Thus the ordinary value $g^{\text{add}}(0)$ is a centering boundary, not a physical coefficient at an allowed zero residue. The punctured identity (84) records the same quantity as an explicit correction that is rank one in the complementary s -coordinate while keeping the core value $g(0) = 0$. It is an exact reorganization, not an additional cancellation theorem.

The scope is therefore precise. The physical zero coordinate faces are handled by [theorem 3.1](#); punctured centering removes the artificial zero value and complete constant cycles; and [proposition 4.2](#) controls the exposed clean Möbius mean. The remaining nonconstant weighted core, the CRT aliases, and the terminal $s = 1$ layer are not estimated here. In particular, this section does not prove the full coefficient gate.

5 Punctured two-character dispersion

The complete complementary-factor trace from TPC-36 vanishes after the right coefficient is centered on the punctured line $\mathbb{F}_q^\times$. This does not by itself estimate an incomplete or weighted complementary-factor sum. In this section we compute that weighted sum exactly. The resulting Jacobi matrix is, apart from one rank-one direction, a scaled unitary matrix. Thus the Jacobi expansion does not manufacture cancellation: it transfers a concrete Hadamard energy between two multiplicative-character coordinates.

5.1 Conventions and the weighted transform

Let $q > 2$ be prime, put

$$G := \mathbb{F}_q^\times, \quad n := |G| = q-1, \quad (114)$$

and write $\widehat{G}$ for the group of multiplicative characters of G . Every character, including the principal character $\mathbf{1}$, is extended to $\mathbb{F}_q$ by the value zero at zero. For $v : G \rightarrow \mathbb{C}$, use the unnormalized Mellin transform

$$\widehat{v}(\psi) := \sum_{x \in G} v(x) \overline{\psi(x)}, \quad v(x) = \frac{1}{n} \sum_{\psi \in \widehat{G}} \widehat{v}(\psi) \psi(x). \quad (115)$$

Thus

$$\sum_{x \in G} |v(x)|^2 = \frac{1}{n} \sum_{\psi \in \widehat{G}} |\widehat{v}(\psi)|^2. \quad (116)$$

Fix $h \in G$. Let $f, g : \mathbb{F}_q \rightarrow \mathbb{C}$ satisfy

$$\sum_{D \in \mathbb{F}_q} f(D) = 0, \quad g(0) = 0, \quad \sum_{U \in G} g(U) = 0. \quad (117)$$

The last two conditions imply additive centering of g on $\mathbb{F}_q$, but retain the value zero at the puncture. For $a, s \in G$, define

$$R_s(a) := \sum_{D \in \mathbb{F}_q} f(D)g(s^{-1}(aD + h)), \quad \hat{R}_s(\chi) := \sum_{a \in G} R_s(a)\overline{\chi(a)}. \quad (118)$$

Given an arbitrary weight $w : G \rightarrow \mathbb{C}$, put

$$W_\psi := \hat{w}(\psi), \quad G_\psi := \hat{g}(\psi), \quad F_\chi := \sum_{D \in G} f(D)\chi(D), \quad (119)$$

$$H_\chi := \sum_{s \in G} w(s)\hat{R}_s(\chi). \quad (120)$$

In particular,

$$G_1 = 0, \quad F_1 = -f(0). \quad (121)$$

For characters $\alpha, \beta \in \hat{G}$, let

$$J(\alpha, \beta) := \sum_{x \in \mathbb{F}_q} \alpha(x)\beta(1-x) \quad (122)$$

be the Jacobi sum with the same zero-extension convention; see, for example, [2, Chapter 8] for the classical identities used below.

Theorem 5.1 (Weighted two-character formula). *For every $\chi \in \hat{G}$, define*

$$C_\chi := \frac{\overline{\chi(-h)}}{n} \sum_{\psi \neq 1} W_\psi G_\psi \psi(h) J(\psi, \bar{\chi}). \quad (123)$$

Equivalently,

$$C_\chi = \sum_{s, U \in G} w(s)g(U)\overline{\chi(sU - h)}. \quad (124)$$

Then

$$\boxed{H_\chi = F_\chi C_\chi \quad (\chi \neq 1), \quad H_1 = qF_1 C_1.} \quad (125)$$

Moreover,

$$\boxed{C_1 = -\frac{1}{n} \sum_{\psi \neq 1} W_\psi G_\psi \psi(h).} \quad (126)$$

Thus punctured centering removes the principal ψ -coordinate, but an incomplete weight can still leave a principal χ -coordinate.

Proof. First take $\chi \neq 1$. The $D = 0$ part of $\hat{R}_s(\chi)$ vanishes by character orthogonality. For $D \neq 0$, set $U = s^{-1}(aD + h)$. Since the characters vanish at zero, the omitted value corresponding to $a = 0$ may be restored, and

$$\hat{R}_s(\chi) = F_\chi \sum_{U \in G} g(U)\overline{\chi(sU - h)}. \quad (127)$$

Insert Mellin inversion for w . For fixed $U \in G$, the change of variables $x = sU/h$ gives

$$\sum_{s \in G} \psi(s)\overline{\chi(sU - h)} = \psi(h)\overline{\psi(U)\chi(-h)} \sum_{x \in \mathbb{F}_q} \psi(x)\overline{\chi(1-x)}$$

$$= \psi(h) \overline{\psi(U)\chi(-h)} J(\psi, \bar{\chi}). \quad (128)$$

This calculation does not use nonprincipality of χ ; together with Mellin inversion, it also proves the equivalence of (123) and (124) for every χ . Summing in U produces G_ψ , and the term $\psi = \mathbf{1}$ vanishes by (121). This proves (123) and the nonprincipal part of (125).

For the principal character, sum $R_s(a)$ over $a \in G$. The $D = 0$ term is $nf(0)g(h/s)$. For $D \neq 0$, the affine argument of g runs through $\mathbb{F}_q \setminus \{h/s\}$; hence its sum is $-g(h/s)$. Since $\sum_{D \neq 0} f(D) = -f(0)$,

$$\widehat{R}_s(\mathbf{1}) = qf(0)g(h/s). \quad (129)$$

On the other hand, $J(\psi, \mathbf{1}) = -1$ for every $\psi \neq \mathbf{1}$. Therefore (123) at $\chi = \mathbf{1}$ gives (126). Multiplicative Fourier inversion also shows that its right side is $-\sum_s w(s)g(h/s)$. Combining this fact with $F_1 = -f(0)$ and (129) proves the principal identity in (125). $\square$

Corollary 5.2 (Exceptional Jacobi channels). *For $\chi \neq \mathbf{1}$,*

$$C_\chi = -\frac{W_\chi G_\chi}{n} + \frac{\overline{\chi(-h)}}{n} \sum_{\psi \neq \mathbf{1}, \chi} W_\psi G_\psi \psi(h) J(\psi, \bar{\chi}). \quad (130)$$

The omitted principal ψ -channel is zero. The diagonal channel uses

$$J(\chi, \bar{\chi}) = -\chi(-1) \quad (131)$$

and has no $q^{1/2}$ factor. If $\psi, \chi, \psi\bar{\chi}$ are all nonprincipal, then

$$J(\psi, \bar{\chi}) = \frac{\tau(\psi)\tau(\bar{\chi})}{\tau(\psi\bar{\chi})}, \quad |J(\psi, \bar{\chi})| = q^{1/2}, \quad (132)$$

where τ denotes the usual Gauss sum.

Proof. The diagonal term in (123) equals

$$\frac{\overline{\chi(-h)}}{n} W_\chi G_\chi \chi(h) (-\chi(-1)) = -\frac{W_\chi G_\chi}{n}.$$

The remaining assertions are the standard exceptional and generic Jacobi-sum evaluations with the zero-extension convention fixed above. $\square$

For later use, the TPC-36 coherent functional now has the exact split

$$\begin{aligned} \mathcal{G}_q(b, f, g; w) &:= \frac{1}{n} \sum_{\chi \in \widehat{G}} |\widehat{b}(\bar{\chi})|^2 |H_\chi|^2 \\ &= \frac{q^2}{n} |\widehat{b}(\mathbf{1})|^2 |F_1|^2 |C_1|^2 \\ &\quad + \frac{1}{n} \sum_{\chi \neq \mathbf{1}} |\widehat{b}(\bar{\chi})|^2 |F_\chi|^2 |C_\chi|^2. \end{aligned} \quad (133)$$

The first line on the right is a genuine exceptional face. It vanishes, for example, if $f(0) = 0$, but it is not removed by $g(0) = 0$ alone.

5.2 The Jacobi matrix is almost unitary

The generic size $q^{1/2}$ of a Jacobi sum is misleading if the ψ -sum is kept coherently. Exact orthogonality determines the whole matrix.

Lemma 5.3 (Full Jacobi orthogonality). *For $\psi, \phi \in \widehat{G}$,*

$$\sum_{\chi \in \widehat{G}} J(\psi, \chi) \overline{J(\phi, \chi)} = n(n\mathbf{1}_{\{\psi=\phi\}} - 1). \quad (134)$$

Proof. Expand both Jacobi sums and use character orthogonality in χ . Only pairs with $1 - x = 1 - y \neq 0$, hence $x = y \neq 1$, survive. Therefore the left side is

$$n \sum_{\substack{x \in G \\ x \neq 1}} \psi(x) \overline{\phi(x)}.$$

The last sum is $n - 1$ when $\psi = \phi$, and -1 otherwise, which proves (134). $\square$

Let $\mathcal{X} := \widehat{G} \setminus \{\mathbf{1}\}$, so $|\mathcal{X}| = q - 2 = n - 1$, and define the square matrix

$$K_{\chi, \psi} := J(\psi, \chi), \quad \chi, \psi \in \mathcal{X}. \quad (135)$$

Theorem 5.4 (Rank-one defect and exact singular values). *With $\mathbf{1}$ now denoting the all-ones vector on $\mathcal{X}$,*

$$\boxed{K^* K = K K^* = n^2 I - q \mathbf{1} \mathbf{1}^*}. \quad (136)$$

Moreover,

$$K \mathbf{1} = \mathbf{1}, \quad K^* \mathbf{1} = \mathbf{1}. \quad (137)$$

Consequently the singular values of K are

$$n \quad \text{with multiplicity } q - 3, \quad 1 \quad \text{with multiplicity } 1. \quad (138)$$

For $q = 3$, only the final singular value is present.

Proof. Restrict (134) to $\psi, \phi \in \mathcal{X}$, and remove the term $\chi = \mathbf{1}$. Since $J(\psi, \mathbf{1}) = -1$ for $\psi \in \mathcal{X}$, the removed outer product is $\mathbf{1} \mathbf{1}^*$. Thus the diagonal entries of $K^* K$ are $n(n - 1) - 1 = n^2 - q$, while the off-diagonal entries are $-n - 1 = -q$. This proves the first identity in (136). The second follows either by the symmetric calculation in the other character variable or from the row and column sums below.

Summing $J(\psi, \chi)$ over all $\psi \in \widehat{G}$ gives zero. Since $J(\mathbf{1}, \chi) = -1$ for nonprincipal χ , each restricted row sum is one. The same argument in χ gives each restricted column sum one, proving (137). On the line spanned by $\mathbf{1}$, the squared singular value is

$$n^2 - q(q - 2) = 1.$$

On its orthogonal complement, (136) is $n^2 I$. This proves (138). $\square$

Put

$$z_\psi := W_\psi G_\psi \psi(h), \quad \psi \in \mathcal{X}. \quad (139)$$

The diagonal phases $\psi(h)$ and $\overline{\chi(-h)}$ are unitary, so theorem 5.4 gives immediately

$$\boxed{\sum_{\chi \neq \mathbf{1}} |C_\chi|^2 = \sum_{\psi \neq \mathbf{1}} |W_\psi G_\psi|^2 - \frac{q}{n^2} \left| \sum_{\psi \neq \mathbf{1}} W_\psi G_\psi \psi(h) \right|^2}. \quad (140)$$

This is the exact average identity behind the two-character route.

5.3 Multiplicative convolution and shifted Parseval

There is a simpler physical-space interpretation of (140). Define the multiplicative convolution on G by

$$c(x) := (w *_\times g)(x) := \sum_{s \in G} w(s)g(x/s), \quad x \in G, \quad (141)$$

and extend c to zero by $c(0) = 0$. Then

$$\widehat{c}(\psi) = W_\psi G_\psi, \quad \sum_{x \in G} c(x) = 0. \quad (142)$$

Theorem 5.5 (Shifted multiplicative Parseval). *For every $\chi \in \widehat{G}$,*

$$\boxed{C_\chi = \sum_{x \in G} c(x) \overline{\chi(x-h)}}. \quad (143)$$

In particular,

$$C_1 = -c(h), \quad (144)$$

$$\sum_{\chi \in \widehat{G}} |C_\chi|^2 = n(\|c\|_2^2 - |c(h)|^2), \quad (145)$$

$$\sum_{\chi \neq 1} |C_\chi|^2 = \boxed{n\|c\|_2^2 - q|c(h)|^2}. \quad (146)$$

Equivalently, if

$$E_{w,g} := \sum_{\psi \neq 1} |W_\psi G_\psi|^2, \quad (147)$$

then

$$\boxed{\frac{E_{w,g}}{n^2} \leq \sum_{\chi \neq 1} |C_\chi|^2 \leq E_{w,g}}. \quad (148)$$

The lower constant is sharp for every $q > 2$; the upper constant is sharp for $q > 3$.

Proof. Regroup the defining sum according to $x = sU$:

$$\sum_{s, U \in G} w(s)g(U) \overline{\chi(sU-h)} = \sum_{x \in G} c(x) \overline{\chi(x-h)}.$$

This is (143). The principal character deletes the single value $x = h$; since $\sum_x c(x) = 0$, this gives (144). Apply Mellin Parseval to the function $y \mapsto c(y+h)$ on G . Its squared norm is $\|c\|_2^2 - |c(h)|^2$, because $c(0) = 0$. This proves (145); subtracting $|C_1|^2$ proves (146).

By (142) and Mellin Parseval,

$$E_{w,g} = n\|c\|_2^2. \quad (149)$$

The upper bound in (148) follows at once. For the lower bound, zero-sum Cauchy gives

$$|c(h)|^2 \leq \frac{n-1}{n} \|c\|_2^2.$$

Insert this estimate into (146) and use (149). Equality in the lower bound occurs when c is constant off h and its value at h is the negative of the sum of the other $n-1$ values. For $q > 3$, a nonzero centered c with $c(h) = 0$ gives equality in the upper bound. Both examples belong to the present class by taking $w = \mathbf{1}_{\{1\}}$ and $g = c$. $\square$

The ratio between the two sides of (148) is not a technical defect. It is the condition number of deleting one translated coordinate from a zero-sum vector. Away from that one direction, the two-character Jacobi map is an isometry after division by $q-1$.

5.4 Interval weights and one full average saving

Let I be an interval of integers, and let its nonzero residue multiplicity be

$$w_I(r) := \#\{s \in I : s \equiv r \pmod{q}\}, \quad r \in G. \quad (150)$$

For $\psi \neq \mathbf{1}$, complete periods cancel and

$$W_{I,\psi} = \sum_{\substack{s \in I \\ q \nmid s}} \overline{\psi(s)}. \quad (151)$$

The classical Pólya–Vinogradov inequality [4, Chapter 9] therefore gives, uniformly in the position and length of I ,

$$B_I := \max_{\psi \neq \mathbf{1}} |W_{I,\psi}| \ll q^{1/2} \log q. \quad (152)$$

Corollary 5.6 (Interval-weight average). *For every punctured-centered g ,*

$$\boxed{\frac{1}{n} \sum_{\chi \neq \mathbf{1}} |C_\chi|^2 \leq B_I^2 \|g\|_2^2 \ll q(\log q)^2 \|g\|_2^2.} \quad (153)$$

More generally, B_I in the first bound may be replaced by any available bound for the nonprincipal Mellin transforms of the actual weight.

Proof. By (147),

$$E_{w_I,g} \leq B_I^2 \sum_{\psi \neq \mathbf{1}} |G_\psi|^2 = n B_I^2 \|g\|_2^2.$$

Use the upper bound in (148) and divide by n . □

A pointwise Cauchy estimate in the incomplete s -variable can lose q in energy after the interval is reduced to a nonconstant remainder modulo q . The averaged estimate (153) removes that full factor, up to logarithms. It is only an unweighted average in χ ; transferring it to the physical coefficient direction requires spectral information about the outer row coefficient.

5.5 Outer Mellin flatness and the principal boundary

For $b : G \rightarrow \mathbb{C}$, define the product–slope operator

$$(Tb)(j, s) := \sum_{\ell \in G} b(\ell) R_s(\ell j), \quad j, s \in G, \quad (154)$$

and put

$$a(t) := (b *_{\times} f)(t) := \sum_{\ell \in G} b(\ell) f(t/\ell). \quad (155)$$

Its transform with the opposite conjugation is

$$\tilde{a}(\chi) := \sum_{t \in G} a(t) \chi(t) = \widehat{b}(\overline{\chi}) F_\chi. \quad (156)$$

Put

$$B_0 := \sum_{\ell \in G} b(\ell), \quad r_0 := B_0 f(0) c(h). \quad (157)$$

Since f is centered on $\mathbb{F}_q$,

$$\sum_{t \in G} a(t) = -B_0 f(0). \quad (158)$$

The weighted physical output is therefore

$$S_h(j) := \sum_{s \in G} w(s)(Tb)(j, s) = r_0 + \sum_{t \in G} a(t)c(tj + h). \quad (159)$$

The first term is the $D = 0$ contribution. It is generally nonzero for the physical ordinary-centered f , even though the ultra coefficient g is punctured.

For $a \neq 0$, define its outer Mellin-flatness constant by

$$\mathfrak{K}_{\text{Mell}}(a) := \frac{\max_{\chi \neq \mathbf{1}} |\tilde{a}(\chi)|^2}{\|a\|_2^2}. \quad (160)$$

Theorem 5.7 (Outer spectrum with its principal boundary). *Under (117),*

$$\sum_{j \in G} |S_h(j)|^2 = \frac{q^2}{n} |r_0|^2 + \frac{1}{n} \sum_{\chi \neq \mathbf{1}} |\tilde{a}(\chi)|^2 |C_\chi|^2. \quad (161)$$

If $a \neq 0$, then

$$\sum_{j \in G} |S_h(j)|^2 \leq \frac{q^2}{n} |r_0|^2 + \mathfrak{K}_{\text{Mell}}(a) \left(\max_{\psi \neq \mathbf{1}} |W_\psi|^2 \right) \|a\|_2^2 \|g\|_2^2. \quad (162)$$

In particular, for an interval weight,

$$\sum_{j \in G} |S_h(j)|^2 \ll q |r_0|^2 + \mathfrak{K}_{\text{Mell}}(a) q (\log q)^2 \|a\|_2^2 \|g\|_2^2. \quad (163)$$

Thus a bound $\mathfrak{K}_{\text{Mell}}(a) = X^{o(1)}$ reaches the complete projective diagonal scale for the nonprincipal outer spectrum, up to soft factors, without a pointwise estimate for every twisted Möbius sum. The principal boundary r_0 must be controlled separately.

Proof. The term $D = 0$ in the definition of R_s gives r_0 . On $D \in G$, put $t = \ell D$; this proves (159). Taking its Mellin transform in j , the nonprincipal coordinate is $\tilde{a}(\chi)C_\chi$. For the principal coordinate, use $C_1 = -c(h)$ and $\tilde{a}(\mathbf{1}) = -B_0 f(0)$. The affine-correlation part therefore contributes r_0 , while the constant term contributes nr_0 . Hence

$$\hat{S}_h(\mathbf{1}) = qr_0, \quad \hat{S}_h(\chi) = \tilde{a}(\chi)C_\chi \quad (\chi \neq \mathbf{1}).$$

Mellin Parseval proves (161).

Bound the first factor by $\mathfrak{K}_{\text{Mell}}(a)\|a\|_2^2$, apply the upper bound in (148), and then use

$$E_{w,g} \leq \left(\max_{\psi \neq \mathbf{1}} |W_\psi|^2 \right) n \|g\|_2^2.$$

This proves (162); (163) follows from (152). $\square$

The flatness hypothesis is sufficient for the nonprincipal part and is not asserted for the physical row coefficient. A weaker weighted overlap estimate between $|\tilde{a}(\chi)|^2$ and $|C_\chi|^2$ would also suffice and is the more flexible target. Neither form estimates r_0 ; omitting that explicit principal term would reinstate the same residue-boundary error that punctured centering was designed to expose.

5.6 A punctured sharp example

The saturation example in TPC-36 used a nonzero value at the residue zero. Puncturing removes that particular example, but not the abstract two-character obstruction.

Proposition 5.8 (Single-character punctured witness). *Let $q > 3$, let $\eta \neq \mathbf{1}$, and set*

$$g(0) = 0, \quad g(U) = \eta(U) \quad (U \in G). \quad (164)$$

Then g is punctured-centered and

$$G_\psi = n \mathbf{1}_{\{\psi=\eta\}}. \quad (165)$$

For every weight w ,

$$C_\chi = \overline{\chi(-h)} W_\eta \eta(h) J(\eta, \bar{\chi}). \quad (166)$$

Consequently

$$|C_\chi| = \begin{cases} |W_\eta|, & \chi = \mathbf{1} \text{ or } \chi = \eta, \\ q^{1/2} |W_\eta|, & \chi \neq \mathbf{1}, \eta. \end{cases} \quad (167)$$

If $w = \mathbf{1}_I$ for a subset $I \subset G$ of size r , then

$$\sum_{\eta \neq \mathbf{1}} |W_\eta|^2 = r(n-r). \quad (168)$$

Thus, if both r and $n-r$ are $\gg q$, there is a nonprincipal η with $|W_\eta| \gg q^{1/2}$, and the generic outputs in (167) have size $\gg q$.

Proof. Character orthogonality gives (165); inserting it into (123) proves (166). The exceptional and generic Jacobi evaluations give (167). Finally, Mellin Parseval gives

$$\sum_{\eta \in \widehat{G}} |W_\eta|^2 = nr, \quad |W_1|^2 = r^2,$$

which proves (168) and the final assertion. $\square$

This witness is not a model for the actual Möbius coefficient. It proves only that punctured centering and interval support, without coefficient-specific information, cannot imply a uniformly small weighted coherent spectrum.

5.7 The actual Möbius Hadamard energy

The physical zero coordinate is first removed by [theorem 3.1](#); the raw-to-punctured rank-one correction and its clean Möbius mean are then handled by [propositions 4.1](#) and [4.2](#). We analyze only the remaining punctured core. For one physical dyadic block, let

$$a_u = -\mu(u) \log u \vartheta(u/U), \quad \tilde{g}(r) = \sum_{\substack{u \in \mathcal{U} \\ u \equiv r \pmod{q}}} a_u. \quad (169)$$

After punctured centering, every $\psi \neq \mathbf{1}$ satisfies the exact identity

$$G_\psi = \sum_{\substack{u \in \mathcal{U} \\ (u,q)=1}} -\mu(u) \log u \vartheta(u/U) \overline{\psi(u)}. \quad (170)$$

Punctured centering changes only the principal transform; it supplies no pointwise estimate for (170). It does, however, give a completely unconditional rms envelope. Put

$$\nu_{\mathcal{U},q}^* := \max_{r \in G} \# \{u \in \mathcal{U} : u \equiv r \pmod{q}\}. \quad (171)$$

Orthogonal projection onto the punctured zero-sum subspace and Cauchy–Schwarz in each residue class give

$$\|g\|_2^2 \leq \nu_{\mathcal{U},q}^* \sum_{u \in \mathcal{U}} |a_u|^2. \quad (172)$$

For an interval $\mathcal{U}$, $\nu_{\mathcal{U},q}^* \leq 1 + |\mathcal{U}|/q$. Hence [corollary 5.6](#) yields

$$\boxed{\frac{1}{n} \sum_{\chi \neq \mathbf{1}} |C_\chi|^2 \ll q(\log q)^2 \left(1 + \frac{|\mathcal{U}|}{q}\right) \sum_{u \in \mathcal{U}} |a_u|^2.} \quad (173)$$

This bound uses no Möbius cancellation and is correspondingly only an envelope; the weighted outer spectrum can still select an exceptional output character.

For integer weights ω_s on a complementary interval, the exact Hadamard energy is therefore

$$\begin{aligned} E_{\mathcal{I},\mu}(q) &:= \sum_{\psi \neq \mathbf{1}} \left| \sum_{\substack{s \in \mathcal{I} \\ (s,q)=1}} \omega_s \overline{\psi(s)} \right|^2 \\ &\quad \times \left| \sum_{\substack{u \in \mathcal{U} \\ (u,q)=1}} -\mu(u) \log u \, \vartheta(u/U) \overline{\psi(u)} \right|^2. \end{aligned} \quad (174)$$

Equivalently, if

$$c_q(x) := \sum_{\substack{s \in \mathcal{I}, u \in \mathcal{U} \\ su \equiv x \pmod{q}}} \omega_s a_u, \quad x \in G, \quad (175)$$

and $\bar{c}_q = n^{-1} \sum_x c_q(x)$, character orthogonality gives

$$\boxed{E_{\mathcal{I},\mu}(q) = n \sum_{x \in G} |c_q(x) - \bar{c}_q|^2.} \quad (176)$$

Thus Route B asks for a residue-variance or spectral-overlap theorem for the actual coefficient, rather than for a formal saving from the Jacobi kernel.

On an unrestricted divisor convolution,

$$\sum_{u|N} -\mu(u) \log u = \Lambda(N). \quad (177)$$

On the physical target range, the restrictions $T < u \leq U_0$ give precisely the ultra increment used in the inherited orbit slice. After reduction modulo q , [\(176\)](#) also contains the integer aliases folded into each residue. Controlling this variance at the required scale is consequently arithmetic information about the truncated von-Mangoldt coefficient; it is not a consequence of puncturing, Pólya–Vinogradov, or Jacobi orthogonality alone.

We make no claim here that the physical outer convolution satisfies $\mathfrak{K}_{\text{Mell}}(a) = X^{o(1)}$, that [\(174\)](#) has the diagonal-size bound, or that the principal and residue-zero corrections are negligible on every reassembly layer. The results above neither breach the parity barrier nor prove a twin-prime correlation estimate. They identify the exact two-character transform, remove one spurious uniform Cauchy loss under a clear flatness hypothesis, and isolate the coefficient-specific theorem still required.

6 Three-modulus face calculus and determinant-first reassembly

After the physical zero-coordinate excision of [section 3](#), the target N and both factors in $su = N$ are units modulo each auxiliary prime. This section keeps three orbit-scale primes separate and computes all eight multiplicative character faces exactly. The calculation has two consequences. First, the seven proper-coordinate faces have one extra power of J^{-1} in quadratic mass. Second, grouping the completed Cartesian block by the integer determinant $F = su - N$, rather than completing s or u separately, removes the apparent incomplete-box boundary loss. Neither statement is a bound for the physical coherent row energy; the precise remaining Bessel-type input is isolated at the end.

6.1 The eight faces on the correct finite-field domain

Let $q_1, q_2, q_3 > 2$ be distinct primes, put

$$n_i = q_i - 1, \quad p_i = \frac{1}{n_i}, \quad M = q_1 q_2 q_3, \quad (178)$$

and fix an integer N satisfying

$$(N, M) = 1. \quad (179)$$

For $F \in \mathbb{Z}/q_i\mathbb{Z}$, define

$$d_i(F) := \mathbf{1}_{F=0}, \quad (180)$$

$$e_i(F) := \mathbf{1}_{F \neq -N}, \quad (181)$$

$$m_i(F) := p_i e_i(F), \quad c_i(F) := d_i(F) - m_i(F). \quad (182)$$

The exclusion $F = -N$ in e_i is essential. Indeed, if $F = su - N$, then $F = -N \pmod{q_i}$ is exactly the residue on which $su = 0 \pmod{q_i}$, so every multiplicative character extended by zero must vanish there.

To make the character interpretation explicit, write

$$x_i(F) := (N + F)N^{-1} \pmod{q_i}. \quad (183)$$

Every multiplicative character of $G_i := \mathbb{F}_{q_i}^\times$ is extended by zero at the origin. The orthogonality relation on G_i gives

$$d_i(F) = \frac{1}{n_i} \sum_{\chi_i} \chi_i(x_i(F)), \quad (184)$$

$$m_i(F) = \frac{1}{n_i} \mathbf{1}_{G_i}(x_i(F)), \quad (185)$$

$$c_i(F) = \frac{1}{n_i} \sum_{\chi_i \neq \mathbf{1}} \chi_i(x_i(F)). \quad (186)$$

Thus m_i is the principal multiplicative face and c_i is the aggregate nonprincipal face. In particular,

$$d_i = m_i + c_i \quad (187)$$

holds also at $x_i = 0$, where all three terms vanish.

For $S \subseteq [3] := \{1, 2, 3\}$, define the corresponding face on $\mathbb{Z}/M\mathbb{Z}$ by

$$K_S(F) := \prod_{i \in S} c_i(F) \prod_{i \notin S} m_i(F). \quad (188)$$

The CRT identifies $F \pmod{M}$ with its three coordinates. Therefore (187) gives the exact eight-face identity

$$\mathbf{1}_{M|F} = \prod_{i=1}^3 d_i(F) = \sum_{S \subseteq [3]} K_S(F). \quad (189)$$

All norms below are unnormalized counting norms on $\mathbb{Z}/M\mathbb{Z}$.

Theorem 6.1 (Orthogonal face calculus). *The eight functions K_S , $S \subseteq [3]$, are pairwise orthogonal in $\ell^2(\mathbb{Z}/M\mathbb{Z})$. For every $S \subseteq [3]$,*

$$\|K_S\|_2^2 = \prod_{i \in S} (1 - p_i) \prod_{i \notin S} p_i, \quad (190)$$

$$\|K_S\|_1 = 2^{|S|} \prod_{i \in S} (1 - p_i). \quad (191)$$

Moreover, at the joint residue $F = 0 \pmod{M}$,

$$K_S(0) = \|K_S\|_2^2. \quad (192)$$

Consequently the squared norms of the empty, singleton, double, and full faces have respective scales

$$J^{-3}, \quad J^{-2}, \quad J^{-1}, \quad 1 \quad (193)$$

when $q_i \asymp J$.

Proof. In one coordinate,

$$\sum_{F \pmod{q_i}} c_i(F) = 0, \quad \langle c_i, m_i \rangle = 0. \quad (194)$$

There are n_i residues on which $e_i = 1$. Hence

$$\|m_i\|_2^2 = n_i p_i^2 = p_i, \quad \|m_i\|_1 = n_i p_i = 1, \quad (195)$$

$$\|c_i\|_2^2 = (1 - p_i)^2 + (n_i - 1)p_i^2 = 1 - p_i, \quad \|c_i\|_1 = (1 - p_i) + (n_i - 1)p_i = 2(1 - p_i). \quad (196)$$

Tensoring through the CRT proves orthogonality and (190)–(191). At $F = 0$, every coordinate equals the distinguished unit residue, so

$$c_i(0) = 1 - p_i, \quad m_i(0) = p_i, \quad (197)$$

which proves (192). The scale table follows from $p_i \asymp J^{-1}$. $\square$

6.2 The coherent union of all proper faces

Put

$$A := \prod_{i=1}^3 (1 - p_i), \quad C(F) := K_{[3]}(F) = \prod_{i=1}^3 c_i(F), \quad (198)$$

and combine the seven proper faces before taking absolute values:

$$P(F) := \sum_{S \subsetneq [3]} K_S(F). \quad (199)$$

By (189),

$$P(F) = \mathbf{1}_{M|F} - C(F). \quad (200)$$

In particular, on every nonjoint residue the proper union and full face cancel exactly.

Theorem 6.2 (Exact norms of the proper union). *With the preceding definitions,*

$$\boxed{\|P\|_\infty = 1 - A,} \quad (201)$$

$$\boxed{\|P\|_2^2 = 1 - A,} \quad (202)$$

$$\boxed{\|P\|_1 = 1 + 6A.} \quad (203)$$

The full face satisfies

$$\|C\|_2^2 = A, \quad \|C\|_1 = 8A. \quad (204)$$

Thus, for $q_i \asymp J$,

$$1 - A \asymp J^{-1}. \quad (205)$$

Proof. The full-face formulas are the $S = [3]$ case of [theorem 6.1](#). Orthogonality of all eight faces and [\(189\)](#) give

$$\|P\|_2^2 = \sum_{S \subsetneq [3]} \|K_S\|_2^2 = 1 - A. \quad (206)$$

At the joint residue, $P(0) = 1 - A$. Away from the joint residue, either one coordinate is the excluded zero residue, in which case $P = C = 0$, or [\(200\)](#) gives $P = -C$. If Z is the nonempty set of failed congruence coordinates at such a residue, then

$$|C(F)| = \prod_{i \in Z} p_i \prod_{i \notin Z} (1 - p_i) \leq \max_i p_i \leq 1 - A. \quad (207)$$

This proves [\(201\)](#).

Finally, the joint residue contributes $1 - A$ to $\|P\|_1$, while the nonjoint residues contribute

$$\|C\|_1 - |C(0)| = 8A - A = 7A. \quad (208)$$

Their sum is $1 + 6A$, proving [\(203\)](#). The final comparison follows from fixedly many $p_i \asymp J^{-1}$. $\square$

The same identities locate the quadratic mass exactly.

Corollary 6.3 (Joint-alias and ghost-face energy). *On one period modulo M ,*

$$|C(0)|^2 = A^2, \quad |P(0)|^2 = (1 - A)^2, \quad (209)$$

$$\sum_{\substack{F \bmod M \\ F \neq 0}} |C(F)|^2 = A(1 - A), \quad \sum_{\substack{F \bmod M \\ F \neq 0}} |P(F)|^2 = A(1 - A). \quad (210)$$

Thus the full-face energy is concentrated on joint aliases, whereas the proper-face energy is concentrated on nonjoint residues that cancel the full face in [\(200\)](#).

Proof. The joint values follow from [\(198\)](#) and [\(200\)](#). Subtract them from the total squared norms in [\(202\)](#) and [\(204\)](#). $\square$

6.3 Determinant-first periodization

The finite calculation becomes useful on the physical reciprocal blocks only after choosing the correct integer variable. Fix $0 < N \ll X$ satisfying [\(179\)](#), and let $\mathcal{R} \subset \mathbb{N}^2$ be any set of pairs (s, u) such that

$$|su - N| \leq B, \quad B \ll X. \quad (211)$$

TPC-36 supplies this hypothesis on every reciprocal dyadic block. Put

$$F(s, u) = su - N. \quad (212)$$

Theorem 6.4 (Determinant-first periodization). *Let H be any of the eight faces K_S , the full face C , or the proper union P . For $r \in \{1, 2\}$ and every fixed $\varepsilon > 0$,*

$$\boxed{\sum_{(s,u) \in \mathcal{R}} |H(F(s,u))|^r \ll_\varepsilon X^\varepsilon \left(1 + \frac{B}{M}\right) \sum_{F \bmod M} |H(F)|^r.} \quad (213)$$

The same estimate holds after inserting unary weights of pointwise size $X^{o(1)}$. It is uniform in the shape of the reciprocal rectangle.

Proof. For a fixed integer F , every pair counted on the left satisfies

$$su = N + F. \quad (214)$$

When $N + F > 0$, the number of positive representations is at most $\tau(N + F) \ll_\varepsilon X^\varepsilon$; when $N + F \leq 0$, there are none. Therefore

$$\sum_{(s,u) \in \mathcal{R}} |H(F(s,u))|^r \ll_\varepsilon X^\varepsilon \sum_{|F| \leq B} |H(F)|^r. \quad (215)$$

The function H is periodic modulo M . Partitioning $[-B, B] \cap \mathbb{Z}$ into complete residue blocks and at most two partial blocks gives

$$\sum_{|F| \leq B} |H(F)|^r \ll \left(1 + \frac{B}{M}\right) \sum_{F \bmod M} |H(F)|^r. \quad (216)$$

Combining the last two displays proves the theorem. Pointwise-soft unary weights are absorbed into X^ε . $\square$

The theorem is the reason to group by F before completing either s or u . A one-variable residue count produces an artificial boundary such as $\min\{S, U\}/J$; the determinant fiber has only divisor-bounded multiplicity and sees the exact M -periodic norm.

Corollary 6.5 (Endpoint periodization ledger). *Assume $q_i \asymp J$, $B \ll X$, and put*

$$\mathcal{K}_X := 1 + \frac{X}{J^3}. \quad (217)$$

Then every individual face and the proper union satisfy

$$\sum_{(s,u) \in \mathcal{R}} |K_S(F(s,u))| + \sum_{(s,u) \in \mathcal{R}} |P(F(s,u))| \ll X^{o(1)} \mathcal{K}_X. \quad (218)$$

In quadratic mass,

$$\sum_{(s,u) \in \mathcal{R}} |P(F(s,u))|^2 \ll X^{o(1)} \frac{\mathcal{K}_X}{J}, \quad (219)$$

$$\sum_{(s,u) \in \mathcal{R}} |C(F(s,u))|^2 \ll X^{o(1)} \mathcal{K}_X. \quad (220)$$

At the endpoint (59),

$$\mathcal{K}_X = \frac{X}{J^3} X^{o(1)} = \frac{Q}{J^2} X^{o(1)} = X^{1/400+o(1)}, \quad (221)$$

$$\frac{\mathcal{K}_X}{J} = \frac{X}{J^4} X^{o(1)} = X^{-132/400+o(1)}. \quad (222)$$

Proof. Apply [theorem 6.4](#) with the exact norms in [theorems 6.1](#) and [6.2](#). Since every face has $O(1)$ one-period ℓ^1 norm, the first bound follows. The proper and full ℓ^2 bounds use respectively $1 - A \asymp J^{-1}$ and $A \asymp 1$. The endpoint identities follow from $QJ = X^{1+o(1)}$ and $J = X^{133/400+o(1)}$. $\square$

Remark 6.6 (Quadratic sparsity is not physical closure). Equation [\(222\)](#) is an entrywise/fiberwise quadratic statement. The physical orbit slice sums in s, u , the input rows, and the determinant shifts before taking its square. No inequality in this section transfers [\(222\)](#) through those coherent sums. In particular, it is not a proof of the inherited coefficient gate.

6.4 Aliases, proper-coordinate ghosts, and the terminal fiber

For every integer k , the joint alias $F = kM$ has the same face values as $F = 0$:

$$C(kM) = A, \quad P(kM) = 1 - A. \quad (223)$$

Thus the $O(X/M)$ joint aliases have respective quadratic masses

$$O\left(\frac{X}{M}\right) \quad \text{and} \quad O\left(\frac{X}{MJ^2}\right) \quad (224)$$

on the full and proper faces. The larger proper total in [\(219\)](#) comes from nonjoint residues where $P = -C$.

This cancellation is not optional. For example, work in a terminal-size interval of u 's of length $\asymp X$, set $s = 1$, and suppose $(N, M) = 1$. On the family

$$u = N + kq_1q_2, \quad q_3 \nmid k, \quad q_3 \nmid N + kq_1q_2, \quad (225)$$

the first two congruences hold, the third fails, and all three ratio coordinates are nonzero. There are

$$\asymp \frac{X}{q_1q_2} \asymp \frac{X}{J^2} \quad (226)$$

such points. At each one,

$$C(F) = -(1 - p_1)(1 - p_2)p_3, \quad P(F) = -C(F), \quad (227)$$

so the full face alone has total absolute mass

$$\asymp \frac{X}{J^2} \frac{1}{J} = \frac{X}{J^3}. \quad (228)$$

The original joint CRT delta is zero on the whole family. Dropping the proper faces before their cancellation with C would therefore create a false alias family at exactly the critical three-modulus scale.

On the genuine integer equality $F = 0$, the situation is different. Every proper divisor layer and the terminal layer $s = 1, u = N$ acquire the same scalar face values

$$P(0) = 1 - A \ll J^{-1}, \quad C(0) = A = 1 - O(J^{-1}). \quad (229)$$

Let $V_{\text{eq}}^{\text{abs}}$ denote the sign-blind physical energy of the regular exact-equality kernel. The inherited envelopes give

$$V_{\text{eq}}^{\text{abs}} \ll X^{o(1)} Q^3 J. \quad (230)$$

The exact scalar relation in [\(229\)](#) therefore yields

$$\boxed{V_{\text{eq,proper}} \ll X^{o(1)} J^{-2} Q^3 J = X^{o(1)} \frac{Q^3}{J}.} \quad (231)$$

This closes all proper faces on the actual equality without using a sign estimate. By contrast, the full face is A times the same vector and retains it with asymptotic coefficient one. In particular, the fused terminal Möbius correlation is not weakened by coordinate centering.

Corollary 6.7 (Full-face centering obstruction). *No uniform power saving for vectors supported on $F = 0$ can follow from three-coordinate multiplicative centering alone. More precisely, on that subspace the full-face operator is scalar multiplication by*

$$A = 1 - O(J^{-1}). \quad (232)$$

This is a method obstruction for the CRT-centering route, not a lower bound for the physical Möbius correlation.

6.5 A sharp coherent no-go and the next gate

The L^2 saving for P cannot be upgraded uniformly over all bounded unary coefficients. Retain $q_i \asymp J$ and the endpoint scales. In the same terminal geometry, take $s = 1$, let I be an interval of u 's of length $B \asymp X$, and assume $B/M \rightarrow \infty$. Define the bounded unary weight

$$g_N(u) = \begin{cases} \overline{P(u-N)} / |P(u-N)|, & P(u-N) \neq 0, \\ 0, & P(u-N) = 0. \end{cases} \quad (233)$$

Proposition 6.8 (Sharp quadratic-to-coherent separation). *For the preceding weight,*

$$\left| \sum_{u \in I} g_N(u) P(u-N) \right| \asymp \frac{B}{M}, \quad (234)$$

$$\sum_{u \in I} |P(u-N)|^2 \asymp \frac{B}{MJ}. \quad (235)$$

At the endpoint $B \asymp X$, the ratio of the first quantity to the square root of the second is

$$\boxed{\asymp \sqrt{\mathcal{K}_X J} = \sqrt{\frac{X}{J^2}} = X^{67/400+o(1)}}. \quad (236)$$

Thus no estimate uniform over all bounded unary coefficients can upgrade the displayed periodic L^2 mass to the required coherent scale with only an $X^{o(1)}$ loss.

Proof. The definition of g_N makes the first left side equal to $\sum_{u \in I} |P(u-N)|$. Decompose $I - N$ into complete M -periods and at most two partial periods. The exact one-period norms (203)–(202), together with $B/M \rightarrow \infty$, give (234)–(235). Finally use $\mathcal{K}_X = X/J^3$ and the endpoint value of J . $\square$

The saturating g_N is an artificial periodic coefficient. The proposition does not assert that the physical weight $-\mu(u) \log u$ has this alignment. It proves that avoiding it requires coefficient-specific arithmetic information.

To state the remaining target correctly, the dependence on the moving physical target must be retained. Restore it in the notation by writing $C_N(F)$ and $P_N(F)$. The one-period norms are independent of the unit $N \bmod M$. Therefore, directly from periodicity and the exact norms—without using divisor multiplicity—one has, uniformly for $(N, M) = 1$,

$$\begin{aligned} \sum_{|F| \ll X} |C_N(F)|^2 &\ll \mathcal{K}_X, \\ \sum_{|F| \ll X} |P_N(F)|^2 &\ll \frac{\mathcal{K}_X}{J}. \end{aligned} \quad (237)$$

Let $\mathcal{N}_{\text{reg}}$ denote the targets occurring in one physical packet after the zero-face excision, so every $N \in \mathcal{N}_{\text{reg}}$ satisfies $(N, M) = 1$. Let

$$\mathcal{H}_{\text{phys}} := \ell^2\{(L, \gamma, j)\} \oplus \ell^2\{(R, \alpha, j)\} \quad (238)$$

be the direct sum of the left and right packet output spaces, with the indices restricted to the retained packet. For the unweighted physical output its squared norm is exactly $V_L + V_R$. All dyadic, projective, and reassembly layers belonging to that packet, including their proved $X^{o(1)}$ cost, are included in the outputs below. For a scalar field $\kappa = (\kappa_N(F))_{N \in \mathcal{N}_{\text{reg}}}$, let $\mathcal{Z}_F[\kappa]$ denote the Hilbert-space-valued physical output of the shifted determinant fiber

$$su = \ell dj + h + F. \quad (239)$$

Thus $\mathcal{Z}_F[\kappa] \in \mathcal{H}_{\text{phys}}$, and every atomic summand with target $N = \ell dj + h$ is multiplied by $\kappa_N(F)$ before the coherent row sum. This notation is necessary: the coordinate faces depend on $N \bmod M$ and cannot be pulled out as a single scalar depending only on F .

Conditional input 6.9 (Target-coupled shifted-fiber Bessel input). For the actual row mask and Möbius-derived coefficients,

$$\left\| \sum_{|F| \ll X} \mathcal{Z}_F[\kappa] \right\|_2^2 \ll X^{o(1)} Q^2 J \sup_{N \in \mathcal{N}_{\text{reg}}} \sum_{|F| \ll X} |\kappa_N(F)|^2 \quad (240)$$

for the two fields $\kappa_N = C_N$ and $\kappa_N = P_N$.

This is a sufficient input, not a theorem of the present paper. Under [conditional input 6.9](#), [\(237\)](#) gives

$$Q^2 J \mathcal{K}_X = Q^2 J \frac{Q}{J^2} = \frac{Q^3}{J}, \quad (241)$$

for the full face, and the stronger scale

$$Q^2 J \frac{\mathcal{K}_X}{J} = \frac{Q^3}{J^2}. \quad (242)$$

The conditional input [\(240\)](#) is not a formal consequence of the periodic L^2 calculation. [Proposition 6.8](#) shows that arbitrary bounded unary data can align with the L^1 , rather than L^2 , size of the face. Any proof at the stated physical normalization must therefore exploit the actual row and Möbius signs, or an equivalent dispersion mechanism. There is also a genuine uniformity issue: on the F -fiber the affine shift is

$$h_F = h + F, \quad (243)$$

which may be of size X . Fixed- h primitivity and incidence bounds from the preceding papers cannot simply be invoked uniformly in F . The three-modulus calculation therefore removes the fixed-target proper-coordinate and incomplete-period bookkeeping obstructions, but leaves a precise target-coupled, coefficient-weighted shifted-fiber gate.

7 Endpoint ledger and the relocated coherent-face gates

The three routes above answer different questions and must not be merged into a single cancellation claim. This section records their exact logical output at the endpoint and isolates two sufficient, but presently unproved, continuations.

7.1 What has been removed from the regular problem

Let

$$\mathfrak{G}_X := X^{o(1)} \frac{Q^3}{J} \quad (244)$$

denote the inherited target scale. The reductions proved above have the following consequences.

- (i) By [theorem 3.1](#) and [corollary 3.2](#), every physical row–time pair on which one of the chosen CRT primes divides the target contributes $O(\mathfrak{G}_X)$. Hence the modular analysis may begin with both complementary coordinates in the unit groups.
- (ii) By [proposition 4.1](#), ordinary additive centering can be replaced by punctured centering without hiding a zero-residue coefficient. The difference is an explicit term that is rank one in the complementary s -coordinate and contains the clean scalar β_U ; complete constant residue cycles vanish by [proposition 4.3](#).
- (iii) By [theorems 5.4](#) and [5.5](#), the nonconstant complementary weight is represented exactly by a Hadamard product of two Mellin spectra. The Jacobi transform has only a rank-one defect and therefore supplies no generic cancellation. For an interval, Pólya–Vinogradov removes one uniform Cauchy factor only in the unweighted character average.
- (iv) By [theorems 6.1](#) and [6.2](#), the seven proper coordinate faces of three moduli form one explicit kernel P , with one-period squared mass $\asymp J^{-1}$. Determinant-first grouping gives the fiberwise bound [\(219\)](#), and on the exact equality the proper-face energy is $O(\mathfrak{G}_X)$ by [\(231\)](#).

None of these statements estimates the full regular output after all row, orbit, dyadic, and alias sums are restored. In particular, the fiberwise L^2 estimate for P is not a license to apply Cauchy through a coherent sum; [proposition 6.8](#) quantifies the resulting polynomial loss for arbitrary bounded coefficients.

7.2 The two surviving formulations

The unsuppressed equality face and the coherent off-equality alias reassembly can be attacked in two nonexclusive forms.

Outer spectral overlap. For one completed product–slope block, put $B_0 = \sum_{\ell} b(\ell)$, $c = w *_\times g$, and $r_0 = B_0 f(0) c(h)$. The exact arithmetic object is

$$\mathfrak{S}_q^{\text{full}}(a, w, g) := \frac{q^2}{q-1} |r_0|^2 + \frac{1}{q-1} \sum_{\chi \neq 1} |\tilde{a}(\chi)|^2 |C_\chi(w, g)|^2, \quad (245)$$

where $a = b *_\times f$ is the actual outer row convolution and

$$C_\chi(w, g) = \sum_{x \in \mathbb{F}_q^\times} (w *_\times g)(x) \overline{\chi(x-h)}.$$

[Theorem 5.7](#) bounds the nonprincipal sum if the actual a obeys $\mathfrak{K}_{\text{Mell}}(a) = X^{o(1)}$; it leaves r_0 explicit. That pointwise flatness hypothesis is stronger than necessary: an aggregate estimate for the nonprincipal overlap, averaged over the physical layers, would suffice, provided the principal r_0 -term is also controlled. Neither input is proved here.

Shifted determinant fibers. Keeping the integer determinant F visible gives the alternative target-coupled Hilbert-space input (240). Its normalization is not arbitrary: the direct period masses (237) imply

$$\begin{aligned} \text{full face: } Q^2 J \mathcal{K}_X &= Q^3/J, \\ \text{proper union: } Q^2 J \mathcal{K}_X/J &= Q^3/J^2. \end{aligned} \tag{246}$$

Thus the target-coupled, coefficient-weighted Bessel input at the stated normalization would close the full face and overclose the proper union. The moving shift $h_F = h + F$, and the dependence of the face on $N \bmod M$, prevent one from importing a fixed- h incidence estimate without a new uniformity argument.

These are alternative sufficient inputs for the same desired physical output; they are not asserted to be equivalent term by term. The first asks for avoidance of large Mellin channels after completion; the second asks for almost-orthogonality between integer determinant fibers before completion. Failure of a pointwise flatness bound therefore does not rule out a weighted-overlap or fiberwise theorem.

7.3 A route tree with valid fallback results

The next layer has three mathematically distinct outcomes.

Positive overlap theorem.

Prove the required aggregate nonprincipal overlap bound using the actual row and truncated Möbius coefficients, and separately absorb both the exposed β_U -mean corrections and the principal r_0 -term in (245). This would close the completed finite-field coherent core; integer aliases and all normalization constants would still have to be checked in the physical reassembly.

Positive shifted-fiber theorem.

Prove the target-coupled input (240) uniformly for the moving shifts and actual coefficients. This route directly absorbs the three-modulus alias count at its critical $X^{1/400+o(1)}$ scale.

Scoped obstruction theorem.

Exhibit, within a precisely specified coefficient class, unavoidable concentration in the exceptional Mellin direction or unavoidable alignment between shifted fibers. Such a theorem would rule out that proof mechanism at the claimed normalization, not establish a lower bound for a prime-pair correlation.

The present paper advances the program regardless of which branch is viable: zero faces, artificial centering values, constant residue cycles, and proper equality faces are no longer mixed into the unsuppressed equality-face question. The surviving gates are coefficient-specific and have two exact norm formulations.

7.4 Relation to the parity boundary

The sharp examples in propositions 5.8 and 6.8 show that centering, support restrictions, and periodic L^2 sparsity alone do not force the needed coherent cancellation. This is consistent with the classical parity obstruction for sieve methods [1, 3]. It is not a new formulation or resolution of that obstruction. A future positive theorem would have to use arithmetic information in the actual Möbius-derived coefficient beyond the formal finite-field identities proved here.

8 Exact certificate and claim boundary

The accompanying standard-library program is

`experiments/tpc37_certificate.py.`

It performs deterministic exact-arithmetic checks and writes the canonical JSON archive

`experiments/tpc37_certificate.json.`

The same suite is run by

```
python experiments/tpc37_certificate.py
python -O experiments/tpc37_certificate.py
```

so no check depends on Python `assert` semantics.

8.1 Finite identities checked

The certificate is divided according to the proof architecture:

- (i) row-residue occupancy and the one-hit orbit bound behind [theorem 3.1](#);
- (ii) the punctured-centering identity (84) and complete-cycle annihilation (104);
- (iii) exact Jacobi Gram and energy identities for several small prime fields;
- (iv) the eight-face CRT decomposition, orthogonality, individual norms, and proper-union norms for several triples of primes; and
- (v) determinant periodization and representative joint- and two-coordinate ghost fibers.

All roots of unity used in the Jacobi checks are represented symbolically; the CRT suites use integers and rational numbers. No floating-point tolerance or random sample is used. Finite enumeration verifies the displayed algebra on the archived test families; it is not evidence for an unproved asymptotic cancellation estimate.

8.2 Reproducibility record

The generated JSON records the per-family check counts, byte-for-byte source SHA-256 values, and a certificate digest of the canonical payload. The repository attributes require LF line endings, but the hashing algorithm itself reads the archived bytes without further normalization. For the archived run,

$$\begin{array}{ll} \text{certificate source SHA-256} & \text{3c4565e36931c5fd42d689425cf9ece9b} \\ & \text{ce5d4a376d0657e79cc3f6dab5ccaed,} \end{array} \quad (247)$$

$$\begin{array}{ll} \text{JSON SHA-256} & \text{2ce1c2938868bf07495fb521bb2227bc} \\ & \text{2b2e4a11daeeb5d1d9389b4375fbe0fa,} \end{array} \quad (248)$$

$$\begin{array}{ll} \text{certificate digest} & \text{3aac4337fdfcee4cdb013c73ac5ca56} \\ & \text{2fdcd76a79037e20e4eba6f30ed08efa.} \end{array} \quad (249)$$

These values make changes to the finite certificate auditable without promoting the enumeration to a proof of the analytic theorems.

8.3 Unconditional conclusions

Subject only to the explicit inherited packet hypotheses in [section 2](#), this paper proves:

- (i) physical absorption of one, or any fixed finite number of, zero-coordinate faces at $X^{o(1)}Q^3/J$;
- (ii) the exact punctured-centering identities and corrections that are rank one in the complementary s -coordinate, together with complete-cycle annihilation;

- (iii) the standard superlogarithmic estimate for the exposed clean Möbius mean;
- (iv) the weighted two-character formula, the exact rank-one Jacobi Gram defect, and the shifted multiplicative Parseval identities;
- (v) the interval-weight average bound and the exact outer decomposition into an explicit principal boundary plus a nonprincipal spectrum, with a sufficient flatness criterion for the latter;
- (vi) a punctured sharp example showing that abstract centered data can retain the full coherent character scale;
- (vii) the orthogonal three-modulus face calculus and exact norms of the seven-face proper union;
- (viii) determinant-first ℓ^1 and ℓ^2 periodization;
- (ix) proper-face closure on the exact equality, the full-face scalar obstruction, and the sharp quadratic-to-coherent separation for arbitrary bounded unary weights.

The finite-field flatness theorem is an implication, not a verification of its hypothesis for the physical row coefficient. Similarly, (240) is displayed as a sufficient next input and is not among the proved estimates.

8.4 What is not proved

Neither the manuscript nor the certificate proves:

- (i) the regular or full physical coefficient gate $V_L + V_R \ll X^{o(1)}Q^3/J$;
- (ii) $\mathfrak{K}_{\text{Mell}}(a) = X^{o(1)}$ for the actual outer convolution, the required aggregate nonprincipal spectral-overlap bound, or physical absorption of the exposed β_U -mean corrections and principal r_0 -term;
- (iii) a diagonal-size estimate for the physical Möbius Hadamard energy (174);
- (iv) the target-coupled, coefficient-weighted shifted-fiber Bessel input (240);
- (v) uniform moving-shift incidence control sufficient to reassemble all aliases;
- (vi) a fixed-power saving from the clean Möbius mean;
- (vii) an affine Chowla theorem or a new Möbius-disjointness theorem;
- (viii) a breach of sieve parity or positivity;
- (ix) a Hardy–Littlewood prime-pair asymptotic; or
- (x) infinitely many twin primes.

The precise advance is a reduction theorem. It closes the physical zero faces and the proper faces on the exact equality, and diagonalizes the nonconstant one-modulus boundary. The full nonzero coordinate face is the unique unsuppressed face on the exact equality. Off the equality, coherent reassembly of both the full face and the proper union remains a coefficient-specific problem.

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